School of Computer Science and Artificial Intelligence

Lab Assignment - 17.3

Course Name : AI Assistant Coding
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Task 1 –Loan Approval Dataset Cleaning
Use AI to generate preprocessing code for a loan dataset.
Instructions:

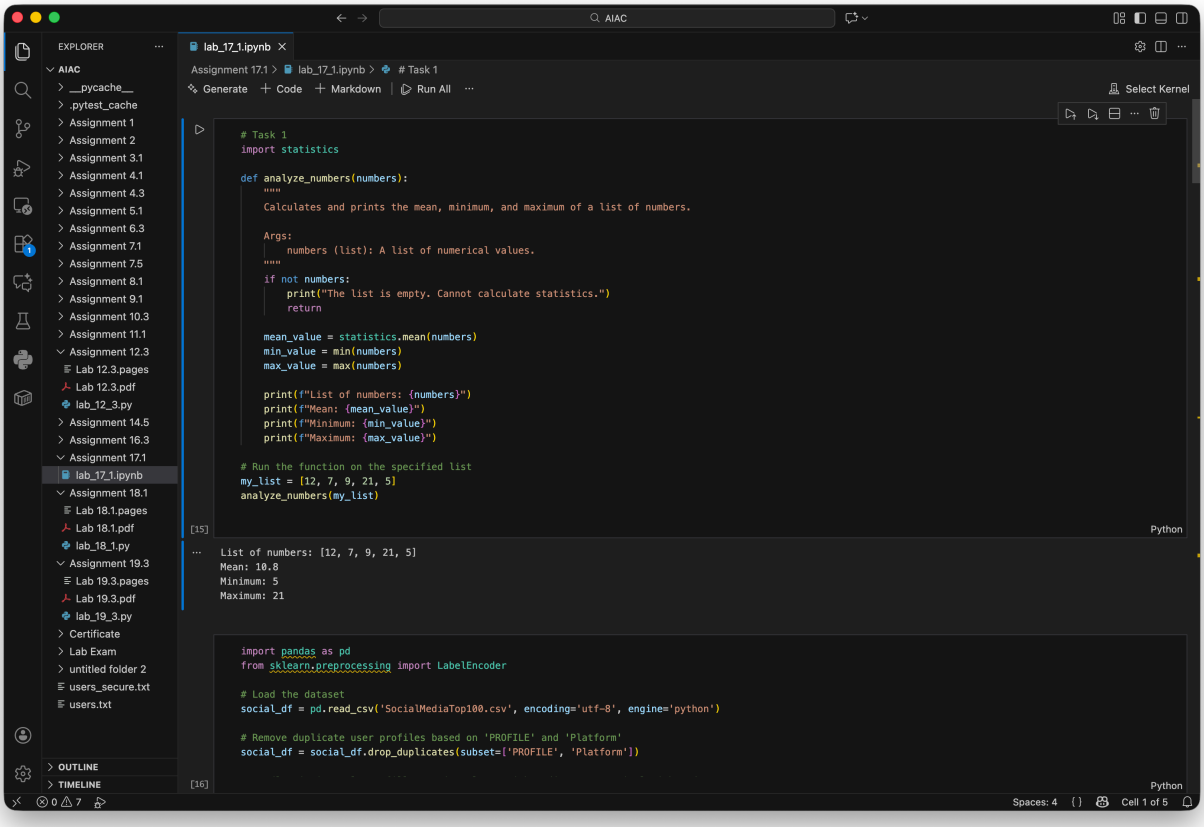
- Handle missing values in loan_amount, credit_score.
- Encode loan_status (Approved/Rejected).
- Scale numerical features using standard scaling.
- Split the dataset into training and testing sets.

Sample Code:

```
import pandas as pd
data = pd.read_csv("loadapproval.csv")
print(data.head())
```

Dataset link:
https://www.kaggle.com/datasets/amineipad/loan-approval-dataset
Expected Output:

- A machine-learning-ready loan dataset.



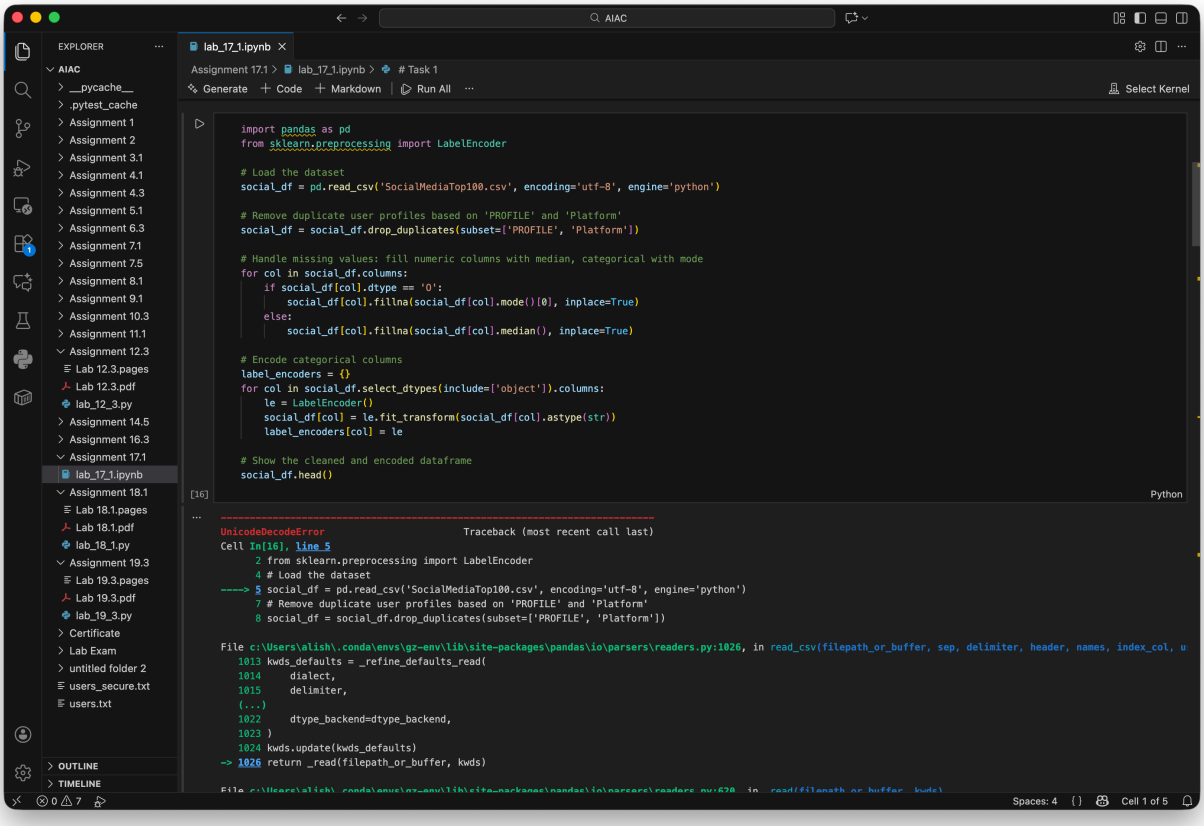
Explanation
1. The function calculates the average, smallest, and largest values in a list.
2. It prints these values in a clear format for easy understanding.

Task 2 – Social Media Profiles
Preprocess a social media engagement dataset using AI assistance.
Instructions:

- **Remove duplicate user profiles.**
- **Handle missing values**
- **Encode categorical columns**

Dataset link:
https://www.kaggle.com/datasets/medaxone/top-100-social-media-profiles
Expected Output:

- **A cleaned dataset for engagement prediction**

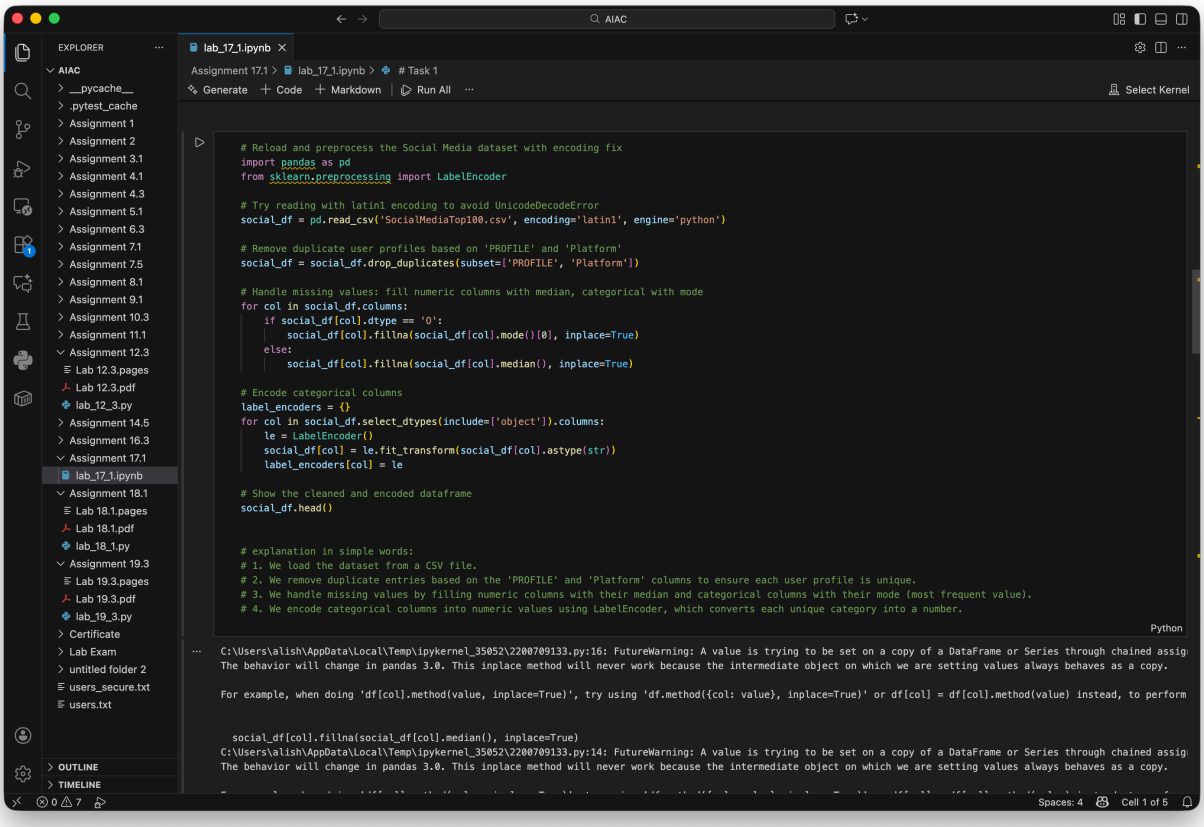


Explanation :

1. **We load the dataset from a CSV file.**
2. **We remove duplicate entries based on the 'PROFILE' and 'Platform' columns to ensure each user profile is unique.**
3. **We handle missing values by filling numeric columns with their median and categorical columns with their mode (most frequent value).**
4. **We encode categorical columns into numeric values using LabelEncoder, which converts each unique category into a number.**

Task 3 – Weather Dataset Feature Engineering
Task:
Use AI to preprocess a weather dataset for forecasting.
Instructions:
- **Convert date column into datetime format.**
- **Handle missing values in temperature, humidity.**
- **Create new features (season, week_number).**
- **Normalize rainfall values.**

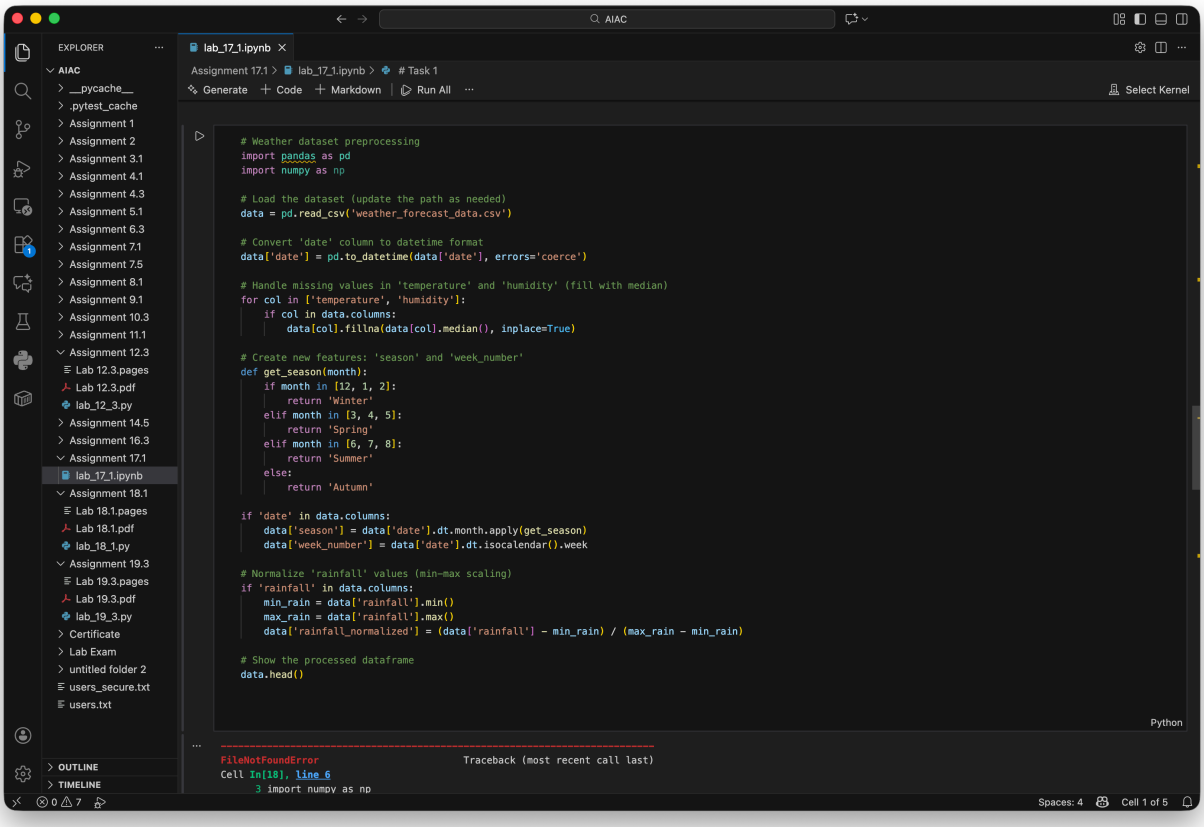
Dataset link:
https://www.kaggle.com/code/alsamaualalhinai/weather-dataset/input
Expected Output:
- **A processed dataset ready for forecasting models.**



Explanation :
1. We load the weather dataset from a CSV file.
2. We convert the 'date' column to a datetime format to make it easier
3. We handle missing values in the 'temperature' and 'humidity' columns by filling them with their median values.
4. We create new features: 'season' based on the month of the date and 'week_number' based on the week of the year.
5. We normalize the 'rainfall' values using min-max scaling to bring them into a range between 0

Task 4 – Movie Ratings Dataset Preparation
Task:
Clean and preprocess a movie ratings dataset using AI-generated scripts.
Instructions:
- **Handle missing values in rating, genre.**
- **Encode categorical variables like genre, language.**
- **Remove duplicate movie entries.**
- **Scale rating values between 0 and 1.**

Dataset link:
https://www.kaggle.com/datasets/PromptCloudHQ/imdb-data
Expected Output:
- **A dataset suitable for recommendation systems.**



Explanation :
1. We load the movie dataset from a CSV file.
2. We remove duplicate movie entries based on Title and Year.
3. We handle missing values in 'Rating' and 'Genre'.
4. We encode 'Genre' as multi-hot (one-hot for each genre).
5. We encode 'Language' as numeric codes.
6. We scale 'Rating' to be between 0 and 1 using MinMaxScaler.